Rakesh Kumar

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Education

Indian Institute of Technology (BHU), Varanasi Dec 2021 – May 2025
*Bachelor of Technology in Computer Science and Engineering (CGPA: **7.93**)* *Varanasi, Uttar Pradesh*

Experience

Samsung R&D Institute India (SRI-Delhi) July 17, 2025 – July 31, 2026
Software Engineer – R&D (Core Platform Software) *Delhi, India*

- Resolved **120+ software defects (Kona)** and submitted **65+ changelists** via Perforce in a multi-threaded **C++** codebase. Investigated and fixed critical memory crashes (SIGABRT/SIGSEGV) and race conditions.
- Owned the **AVControl module**, managing the deployment lifecycle by building and flashing **RPMs** to target TVs. Integrated cross-module configuration APIs to implement core system functionality.
- Engineered an **NLP-based Generative AI** log analyzer utilizing a **multi-agent LangGraph** workflow to dynamically route Kona tickets. Architected a **FAISS-backed RAG** pipeline (**Qwen-3 embeddings**) for historical defect matching and an **MCP server** for selective, token-efficient codebase retrieval.
- Automated thread-level logic and SIGSEGV crash investigations using **LLMs**. Implemented an **Actor-Critic debate pattern** to verify codebase hypotheses, reducing system diagnosis time by **90%**.

Tech Stack: Gen AI, LLMs, RAG, LangGraph, MCP, Agentic AI, Prompt Engineering, C++, OOP, Multithreading, Perforce.

LTIMindtree May 20, 2024 – July 19, 2024
Summer Intern *Delhi, India*

- Performed large-scale web scraping on Zauba Corp, extracting data of **2.7M+ companies** into MySQL.
- Developed a **Flask-based web dashboard** for streamlined database interaction and data visualization.
- Contributed to the development of an **ML-based application** leveraging **Optical Character Recognition (OCR)** techniques to accurately extract transaction details from payment screenshots.

Tech Stack: Python, Flask, BeautifulSoup, Render, GitHub, Cron-job, MySQL, HTML, CSS, JavaScript.

Skills

Programming Languages: C, C++, Python, Java, Dart, JavaScript, SQL.
Tools and Frameworks: Spring Boot, Django, Flask, Node, React, Flutter, MySQL, Supabase, Git, Perforce, REST APIs.
Core Concepts: Gen AI, LLMs, Agentic Workflows, Prompt Engineering, ML, OOP, Multithreading, Web Scraping.

Projects

Proactive AI Personal Assistant | *React*, *FastAPI*, *Supabase* | GitHub| Website Jan 2026 – Apr 2026

- Engineered a full-stack **Generative AI** personal assistant powered by Google Gemini, implementing an **Agentic routing workflow** and structured JSON prompting to extract complex scheduling data with **sub-2.5 second** latency.
- Designed a **100% reliable** background cron system for real-time notifications and spaced-repetition algorithms.

MyMoney | *Personal Finance Tracker* | GitHub | Website Aug 2025 – Nov 2025

- Developed an offline-first personal finance application that allows users to monitor daily expenses, invite peers, and seamlessly manage and settle shared group ledgers.
- Engineered a **Flutter** app with **Firebase** real-time sync, **SQLite (Drift)** local caching, and **OAuth**. Optimized NoSQL schemas via subcollections and batched writes for scalable data reads.

Soulstash | *Movie/Series Tracking Platform* | GitHub | Live July 2024 – Nov 2024

- Built a movie/series discovery platform on a **Microservices** architecture (**React**, **Node.js/Express**, **MongoDB**) with separate gateway, content, user, and collection services, featuring watchlists, custom collections.
- Secured social features using **JWT**, allowing authenticated users to follow peers and publish custom collections.

Honours and Achievements

Competitive Programming

- Maximum rating of **1469 (Specialist)** on Codeforces. Maximum rating of **1494** on CodeChef.
- Secured **Global Rank 50** in CodeChef August Long 2022 Div 4. Rank **1915** in Codeforces Round 852 (Div. 2). Rank **2199** in Educational Codeforces Round 142 (Div. 2).